
CE 203P: Building Materials Lab

Course Code	CE 203P
Course Name	Building Materials Lab
L-T-P-C	0-0-2-1
Prerequisites	None
Intended for	B.Tech. (CE)
Mutual Exclusion	None
Approval	50th BoA (Jul 2023), in the B.Tech Civil Engineering curriculum; 52nd BoA (Nov 2023)

Course Contents

1. Static tension test of mild steel, cast iron and aluminum.
2. Static compression test of fired clay brick, concrete blocks, cubes, cores and cylinders.
3. Measurement of the flexural strength of tiles and concrete.
4. Measurement of the wear/abrasion resistance of tiles and concrete.
5. Water absorption tests for fired clay brick and concrete.
6. Measurement of thermal properties of concrete using hot wire method.
7. Measurement of thermal conductivity of common insulation materials and ghiss using Lees' Disc apparatus.
8. Particle shape and size analyses of aggregates.
9. Specific gravity and water absorption tests for aggregates.
10. Crushing and Impact value tests for coarse aggregates.
11. Fineness and specific gravity tests for cement.
12. Standard consistency and setting time tests for cement.
13. Le Chatelier's and autoclave soundness tests for cement.
14. Use of moisture meter, ultrasonic pulse velocity and rebound hammer tests for the non-destructive assessment of concrete quality .

Text Books

1. Relevant BIS and ASTM standards

References

1. Bahurudeen A., and Moorthi, P.V.P., **Testing of Construction Materials**, CRC Press, 2021.